

Matrix discrepancy and optimization

The sharp Lovász-theta constant for dense random graphs

Difficulty: challenging **Importance:** broadly interesting**Status:** Solution claimed

Rating rationale: A sharp expectation constant requires new random SDP analysis beyond order bounds; it connects matrix optimization with probabilistic combinatorics.

The difficulty and importance ratings above are historical ratings of the original open target.

Full-resolution claim — checked 6 October 2026

Aaron Potechin and Jeff Xu, *Sharp Lovász-Theta Bounds on Random Graphs*, arXiv:2609.30064v1, first submitted 24 September 2026. In §1.1, **Theorem 1.2 and Corollary 1.3** (printed p. 3 of the [full manuscript](#)), the authors claim that $\vartheta(G_n)/\sqrt{n} \rightarrow 1$ in probability. Appendix F (printed pp. 76–77) identifies their formulation with the trace-normalized SDP used in this entry. The v1 corollary prints an exact high-probability lower bound $\vartheta(G_n) \geq \sqrt{n}$. Coauthor Jeff Xu confirmed in correspondence supplied by the maintainer on 8 October 2026 that its intended lower bound is $(1 - o_n(1))\sqrt{n}$. He also confirmed that Claim 4.15’s proof should have a logarithm, rather than n , inside the disputed square root. These corrections leave the leading-constant claim unchanged; a corrected arXiv version was not yet available on 9 October 2026. The [official FOCS 2026 accepted-paper list](#) includes this paper; acceptance is recorded separately from publication in the proceedings.

Correspondence with the expectation target. The claimed probabilistic limit implies the original expectation limit by the following elementary uniform-integrability argument. Let A_n be the ordinary zero-one adjacency matrix and put $B_n = J - I - 2A_n$. For every matrix X feasible in the original SDP, the edge constraints give

$$\langle J, X \rangle = \langle I + B_n, X \rangle \leq 1 + \|B_n\|_2, \quad 0 \leq \frac{\vartheta(G_n)}{\sqrt{n}} \leq \frac{1 + \|B_n\|_2}{\sqrt{n}}.$$

The independent upper-triangular entries of B_n are uniform signs. For a fixed real unit vector x , the sum $x^T B_n x = 2 \sum_{i < j} (B_n)_{ij} x_i x_j$ has squared coefficient sum at most two. The elementary exponential-moment bound for a sum of independent signs therefore gives $\Pr\{|x^T B_n x| > t\} \leq 2e^{-t^2/4}$. A $1/4$ -net of the real unit sphere with at most 9^n points controls the norm of a symmetric matrix within a factor of two. Consequently,

$$\Pr\{\|B_n\|_2 > s\} \leq \min\{1, 2 \cdot 9^n e^{-s^2/16}\}, \quad \mathbb{E}\|B_n\|_2^2 \leq 16(n \log 9 + \log 2 + 1).$$

The moment bound follows by integrating this tail bound. Thus $\sup_n \mathbb{E}[(\vartheta(G_n)/\sqrt{n})^2] < \infty$, so the normalized Lovász numbers are uniformly integrable. Theorem 1.2’s high-probability upper bound therefore gives $\limsup_n \mathbb{E}\vartheta(G_n)/\sqrt{n} \leq 1$. For the lower bound, the deterministic inequality $\vartheta(G)\vartheta(\bar{G}) \geq n$ and the fact that G_n and $\bar{G}_n$ have the same distribution give, by the arithmetic-geometric mean inequality, $\mathbb{E}\vartheta(G_n) \geq \sqrt{n}$ for every n . Together these establish the original expectation limit, conditional on Theorem 1.2. The argument preserves the original SDP, graph ensemble and quantifiers; it does not replace the expectation statement with a probabilistic one.

Evidence, 9 October 2026: an [independent AI-agent review](#) checks the expectation bridge but not the general graph-matrix estimates. Theorem 3.24’s printed $O(1)$ mixed-moment remainder fails; $O(\sqrt{n})$ is sharp, while its equal-shape case retains $O(1)$. Definition 4.3 also makes Lemma 4.11 false as printed. Both failures have plausible local corrections, but the corrected general proof and an MD-01 Lean proof remain unverified. The paper’s “Formal Verification” appendix is a written argument. **Status:** Solution claimed.

Original problem statement

Let G_n be the simple random graph on n labelled vertices in which each unordered pair is an edge independently with probability $1/2$. For a graph G , define its Lovász number by

$$\vartheta(G) = \max\{\langle J, X \rangle : X \in \mathbb{R}^{n \times n}, X = X^T \geq 0, \operatorname{tr} X = 1, X_{ij} = 0 \text{ for } \{i, j\} \in E(G)\},$$

where J is the all-ones matrix and $\langle J, X \rangle = \sum_{i,j} X_{ij}$.

Conjecture. $\lim_{n \rightarrow \infty} \mathbb{E} \vartheta(G_n) / \sqrt{n} = 1$.

This asks for the leading constant of a random semidefinite matrix optimization problem. It is an expectation statement; convergence in probability is not substituted for it.

References

1. A. S. Bandeira, D. Dmitriev, K. Lucca, P. Nizić-Nikolac, and A. Rödder, *Randomstrasse101: Open Problems of 2025*, Entry 9, Conjecture 17 and the preceding SDP definition. [Archived paper](#).
2. A. Potechin and J. Xu, *Sharp Lovász-Theta Bounds on Random Graphs*, arXiv:2609.30064v1, 24 September 2026, §1.1, Theorem 1.2 and Corollary 1.3, printed p. 3; accepted for FOCS 2026.

Status check — 2026-09-10

The archived 2026 paper retains the conjecture. Targeted searches for the Lovász number of dense Erdős–Rényi graphs and a sharp leading constant through September 2026 found no resolution. The adjacent circulant problem uses a different random matrix ensemble and is recorded separately.

Audit update (2026-09-10): Rechecked the 2026 Conjecture 17 and searched for a sharp dense-random-graph theta constant. No resolution of the expectation limit was located. This is a bounded literature check, not a proof that no solution exists.